QUALITY INFORMATION DOCUMENT

Global Ocean Wind Production Centre

WIND_GLO_PHY_L3_NRT_012_002

WIND_GLO_PHY_L3_MY_012_005

Issue: 1.19

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CHANGE RECORD

Issue	Date	§	Description of Change	Author	Validated By
1.0	2013-02-05	All	Creation of the document, Wind QUID split up from the SIW QUID.	Tilly Driesenaar, Abderrahim Bentamy	Ad Stoffelen
1.1	2013-02-28	All	Corrections after QuARG review	Tilly Driesenaar, Abderrahim Bentamy	Ad Stoffelen
1.2	2015-09-01	all	Change format to fit CMEMS graphical rules. Introduce references to CMEMS.	B. Hackett	L. Crosnier
1.3	2015-12-09	all	Change product ID from CLIM to REP	B. Hackett	
1.4	2016-01-19		Updated L3 product info	Jos de Kloe	B. Hackett
1.4.1	2016-04-04	1.3, 2.3	Added EAN numbers	Jos de Kloe	
1.6	2016-07-12	all	Add V2 datasets. Update scatterplot figures. Minor edits.	Jos de Kloe	B. Hackett
1.7	2017-01-16	all	Add L3 REP product	Jos de Kloe	C. Wettre
	2017-03-16	1.3, 2.1, 3.1, 4.1, 7	In response to V3 review: add EAN for curl.div, add figure 1, add equation for artificial input wind field, and one extra reference	Jos de Kloe	Ad Stoffelen
1.8	2017-09-05		Extended ASCAT-A REP time-series and added ERS-1/ERS-2 REP data	M. Belmonte	Ad Stoffelen
1.9	2018-01-26		Rebranded from OSI TAC to WIND TAC	M. Belmonte	Ad Stoffelen
1.10	2018-08-23		Added Oceansat-2 OSCAT REP data and Scatsat-1 OSCAT NRT data. Some editing.	M. Belmonte	Ad Stoffelen
1.11	2019-09-04		Added Metop-C ASCAT data	Rianne Giesen	
1.12	2019-12-16		Textual clarifications after review; added list of data gaps caused by unavailable upstream data in Chapter V.	Rianne Giesen, Ad Stoffelen	Mercator Ocean
1.13	2020-06-24		Figure and EAN updates after review. Name change of L4 REP product and textual clarifications. Updated list of data gaps to include 2019	Abderrahim Bentamy, Isabel Garcia Hermosa, Rianne Giesen	Mercator Ocean
1.14	2020-09-09		Added NRT HY-2B HSCAT datasets. Added information about time series extension and L3 quality flags. Updated list of data gaps.	Rianne Giesen	Ad Stoffelen

Issue	Date	§	Description of Change	Author	Validated By
1.15	2021-09-01		Added NRT HY-2C HSCAT and REP Metop-B ASCAT datasets. Updated information about time series extension. Added recommendation about wind speed calculation from components. Updated list of data gaps.	Rianne Giesen	Ad Stoffelen
1.16	2022-03-30	2.1, 5.1	Changed update frequency of Metop-B ASCAT REP datasets to monthly (005). Metop-A ASCAT datasets no longer updated after end-of-life (Nov 2021). Updated list of data gaps.	Rianne Giesen	Rianne Giesen
1.17	2022-06-13		Added HY-2D HSCAT NRT datasets, updated list of data gaps.	Rianne Giesen	Ad Stoffelen
1.18	2023-09-06	All	Changed product and dataset names, add section about spatial derivatives, quality assessment of winds closer to the coast for 0.125° datasets. Updated list of data gaps.	Rianne Giesen	Ad Stoffelen
1.19	2024-02-01		Added Oceansat-3 OSCAT NRT datasets, updated list of data gaps	Rianne Giesen	Ad Stoffelen

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I. EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of the global ocean near real-time (NRT) and multiyear (MY) reprocessed daily Level-3 (L3) wind products (Table 1).

The input Level-2 (L2) observations are made available by the Ocean and Sea Ice Satellite Application Facility (OSI SAF) services of EUMETSAT and are fully validated against moored buoy winds. The wind variables included are so-called stress-equivalent wind speed, zonal and meridional stress-equivalent wind. Stress-equivalent winds are the closest representation of a scatterometer ocean measurement, which contains no atmospheric information. Buoy winds and European Centre for Medium-range Weather Forecasts (ECMWF) winds are corrected in the OSI SAF for air-mass density and air-mass stability before validation against the L2 scatterometer observations.

The WIND_GLO_PHY_L3_NRT_012_002 product contains daily gridded near-real-time scatterometer wind vector observations, supplemented with wind stress components, wind stress amplitude, and wind and wind stress curl and divergence. Stress-equivalent winds and derivative fields from the operational suite at ECMWF are included for user convenience. The L3 product is produced by the Royal Netherlands Meteorological Institute (KNMI) and disseminated by the Copernicus Marine Service Wind Thematic Assembly Centre (Wind TAC).

The WIND_GLO_PHY_L3_MY_012_005 product contains the same type of observations as the WIND_GLO_PHY_L3_NRT_012_002 product, but it is based on reprocessed L2 climate data records (CDR) as produced by KNMI and EUMETSAT/OSI SAF. The reprocessed product includes ECMWF reanalysis stress-equivalent winds (ERA5; in fact, its short-range forecasts) instead of ECMWF operational model winds.

The scope of this document is not to validate stress-equivalent winds, as this is done at L2 by the OSI SAF. This document addresses the validation of the stress and spatial derivative products, which are all produced under the Copernicus Marine Service umbrella. In addition, the interpolation of the products to L3 and L4 grids is validated.

***Table 1**: Wind TAC products and partner roles*

Product	Product description	Production unit, PU
WIND_GLO_PHY_L3_NRT_012_002	Global ocean wind L3 NRT	SIW-KNMI-DEBILT-NL
WIND_GLO_PHY_L3_MY_012_005	Global ocean wind L3 MY	SIW-KNMI-DEBILT-NL

I.2 Summary of the results

The L3 Wind Product is based on the extensively calibrated, validated and monitored L2 wind products of the EUMETSAT Ocean and Sea Ice Satellite Application Facility (OSI SAF). The verification of the Global Ocean L3 Wind products reported here consists of checking the correct functioning of the gridding procedure by the inspection of individual cases and by buoy collocations for a period of one arbitrary year (2019), and by inserting artificial inputs to the gridding procedure and comparing input and output fields to an exactly known input function. In addition, ocean drag needed to calculate wind stress has been related to stress-equivalent wind in the ECMWF ERA-Interim model.

The verification against moored buoy winds, tested over an entire year of ASCAT 25 km data, shows that the L3 wind product processing works correctly and the level of accuracy is similar to that of the original L2 wind product. The wind vector accuracy is better than 1 m/s (RMS) with a bias of less than 0.2 m/s in wind speed (complying with OSI SAF requirements). The artificial input test shows that errors introduced by the gridding procedure depend on the scale of the wind structures, as expected. For scales of about 300 km and above, the interpolation error is negligible. Relative errors are larger after the interpolation of small-scale structures, where gradients at the smallest 25-km scales may be modified by up to 30%.

The wind stress calculation follows closely the relations used by the ECMWF ERA-Interim wave model, but is still somewhat simplified because sea state dependency is removed. The sea state dependency in the model does not verify with the scatterometer measurements.

I.3 Estimated Accuracy Numbers

The accuracy levels of the Global Ocean L3 wind products are better than 1 m/s in wind component RMS, with a bias of less than 0.2 m/s in wind speed relative to moored buoy winds. This estimate is based on triple collocation of L2 products (e.g., Vogelzang and Stoffelen, 2022) and the additional variance and biases reported here (Table 2). Biases and root-mean-square differences are slightly larger for the L3 products compared to the L2 products, which results from the regridding from the L2 to the L3 grid.

For the wind stress field included in the L3 wind products, we report the difference between the simple linear drag relation as used for the L3 wind product and the drag of the ECMWF ERA-Interim model. This gives differences in wind stress between 5 and 17 percent. This is small compared to stress parameterization uncertainties of about 20-30% (Bourassa et al., 2019).

Derivative fields, such as curl and divergence, may have some uncertainties coming from the original wind or stress fields, but also from the discretization applied during their calculation. The latter depends on the scale of variation as explained in section 4.1, where smaller scales have smaller variances. For a 50 km scale variation and 25 km discretisation, the maximum expected error in these derivatives can be of the order of 50%. For 12.5 km discretisation, this is reduced to around 30%. For 25 km discretisation and a 200 km scale variation it is reduced to about 10%, and for a 400 km scale variation it is reduced to about 2.5%. This interpolation error must be added on top of the error in the original wind or stress field.

***Table 2**: Estimated Accuracy Numbers for the main variables in product WIND_GLO_PHY_L3_NRT_012_002 and WIND_GLO_PHY_L3_MY_012_005 based on a comparison of L3 Metop-A ASCAT observations with moored buoy observations for the year 2019.*

Variable	Metric	Value		Unit
		NRT	MY	
Zonal (eastward) wind	mean bias	0.08	0.08	m s^{-1}
	root-mean-square difference	1.74	1.74	m s^{-1}
Meridional (northward) wind	mean bias	0.06	0.06	m s^{-1}
	root-mean-square difference	1.70	1.70	m s^{-1}
Wind speed	mean bias	-0.06	-0.06	m s^{-1}
	root-mean-square difference	1.10	1.10	m s^{-1}
Wind direction	mean bias	0.8	0.8	degree
	root-mean-square difference	22.6	22.6	degree

II. PRODUCTION SUBSYSTEM DESCRIPTION

The Wind TAC is a multi-mission integration activity. The main tasks to fulfil are:

- provide real-time, delayed mode update operations for L3, and L4 data products for wind.
- provide data for long-term surface wind monitoring data.
- provide quality control, validation and error characterization of data products and services.

Details on the corresponding L2 product, the available period, spatial and temporal resolution for the various datasets can be found in the L3 product user manual (CMEMS-WIND-PUM-012-002-005). The various OSI SAF L2 wind products are visualized and fully described at https://scatterometer.knmi.nl/ . As new satellite instrument data will become available within the OSI SAF, the Copernicus Marine Service L3 functionality will be added and the portfolio extended.

II.1 L3 Global Ocean wind

The global ocean L3 wind product consists of L2 scatterometer wind vectors which are interpolated to a regular lat-lon grid with fixed spacing. The global ocean L3 wind product contains different datasets depending on satellite, grid spacing and whether the data is originating from ascending or descending passes. The L3 grid spacing depends on the resolution of the input L2 product. The input L2 products are operational products from the EUMETSAT OSI SAF produced at KNMI. The production of the Copernicus Marine Service global ocean L3 wind products is also done by the OSI SAF production system at KNMI in The Netherlands (Production Centre). The resulting netCDF product files are then made available through the Copernicus Marine Service Information System.

The various datasets of the global ocean L3 wind product in the Copernicus Marine Service and the L2 products they are based on are listed in Table 3 below.

The L3 NRT product datasets (WIND_GLO_PHY_L3_NRT_012_002) are updated daily. The L3 MY product datasets (WIND_GLO_PHY_L3_MY_012_005) are updated when a new multi-year L2 reprocessing dataset becomes available. The datasets cmems_obs-wind_glo_phy_my_l3-metopb-ascat-XXX-XXXdeg_P1D-i are extended in time every month, such that these datasets cover the period from 2019 up to three months before present. As L2 datasets reprocessed with ERA5 model winds are not yet available for Metop-A and Metop-B ASCAT, the L2 NRT datasets are used as input from 2019 onwards.

***Table 3**: Overview of datasets in the global ocean L3 wind products*

L3 wind dataset	Satellite	Instrument	L2 wind product
cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-des-0.25deg_P1D-i	Metop-A	ASCAT	25 km
cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.125deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-des-0.125deg_P1D-i	Metop-A	ASCAT	12.5 km coastal
cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-des-0.25deg_P1D-i	Metop-B	ASCAT	25 km
cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-asc-0.125deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-des-0.125deg_P1D-i	Metop-B	ASCAT	12.5 km coastal
cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-des-0.25deg_P1D-i	Metop-C	ASCAT	25 km
cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-asc-0.125deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-des-0.125deg_P1D-i	Metop-C	ASCAT	12.5 km coastal
cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-des-0.25deg_P1D-i	ScatSat-1	OSCAT	25 km
cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-des-0.5deg_P1D-i	ScatSat-1	OSCAT	50 km
cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-des-0.25deg_P1D-i	Oceansat-3	OSCAT	25 km
cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-des-0.5deg_P1D-i	Oceansat-3	OSCAT	50 km
cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-des-0.25deg_P1D-i	HY-2B	HSCAT	25 km
cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-des-0.5deg_P1D-i	HY-2B	HSCAT	50 km
cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-des-0.25deg_P1D-i	HY-2C	HSCAT	25 km
cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-des-0.5deg_P1D-i	HY-2C	HSCAT	50 km
cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-des-0.25deg_P1D-i	HY-2D	HSCAT	25 km
cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-des-0.5deg_P1D-i	HY-2D	HSCAT	50 km
cmems_obs-wind_glo_phy_my_l3-metopa-ascat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-metopa-ascat-des-0.25deg_P1D-i	Metop-A	ASCAT	Reprocessed 25 km / 25 km (2019 onwards)
cmems_obs-wind_glo_phy_my_l3-metopa-ascat-asc-0.125deg_P1D-I cmems_obs-wind_glo_phy_my_l3-metopa-ascat-des-0.125deg_P1D-i	Metop-A	ASCAT	Reprocessed 12.5 km coastal / 12.5 km coastal (2019 onwards)
cmems_obs-wind_glo_phy_my_l3-metopb-ascat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-metopb-ascat-des-0.25deg_P1D-i	Metop-B	ASCAT	25 km
cmems_obs-wind_glo_phy_my_l3-metopb-ascat-asc-0.125deg_P1D-i cmems_obs-wind_glo_phy_my_l3-metopb-ascat-des-0.125deg_P1D-i	Metop-B	ASCAT	12.5 km coastal
cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-des-0.5deg_P1D-i	Oceansat-2	OSCAT	Reprocessed 50 km
cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-des-0.25deg_P1D-i	Oceansat-2	OSCAT	Reprocessed 25 km

[continues in the next page]

L3 wind dataset	Satellite	Instrument	L2 wind product
cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-des-0.5deg_P1D-i	QuikSCAT	SeaWinds	Reprocessed 50 km
cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-des-0.25deg_P1D-i	QuikSCAT	SeaWinds	Reprocessed 25 km
cmems_obs-wind_glo_phy_my_l3-ers1-scat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-ers1-scat-des-0.25deg_P1D-i	ERS-1	SCAT	Reprocessed 25 km
cmems_obs-wind_glo_phy_my_l3-ers2-scat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-ers2-scat-des-0.25deg_P1D-i	ERS-2	SCAT	Reprocessed 25 km
cmems_obs-wind_glo_phy_my_l3-ers2-scat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-ers2-scat-des-0.25deg_P1D-i	ERS-2	SCAT	Reprocessed 25 km

The added value of the 0.125° resolution Metop ASCAT datasets is illustrated in Figure 1**Error! Reference source not found.**, clearly showing the increased coverage in coastal regions.

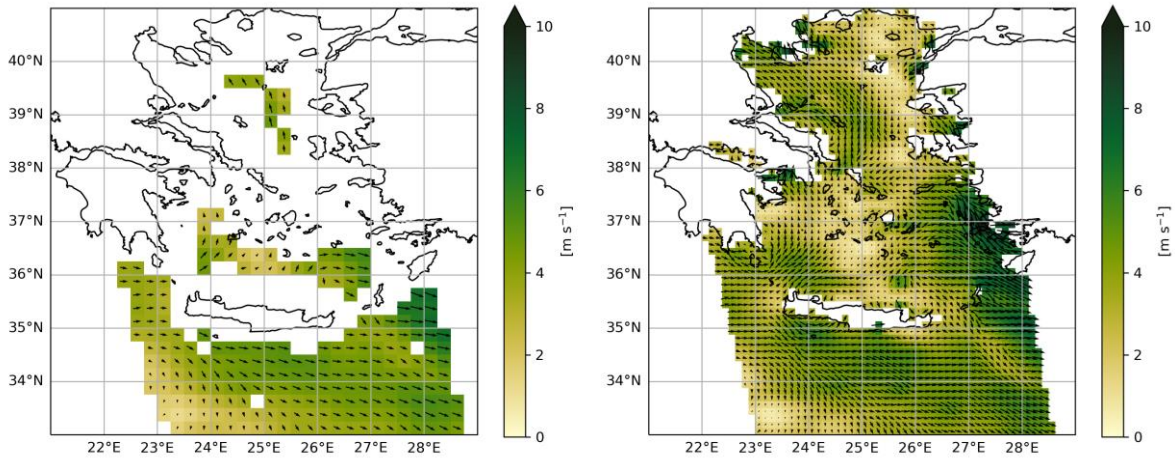

***Figure 1**: Example of the wind field in the 0.25° Metop-C ASCAT dataset (left) and corresponding 0.125° dataset in the eastern part of the Mediterranean on 30 Apr 2023. To illustrate the spatial coverage, all observed wind values are shown, quality flags have not been accounted for.*

The L3 product is derived from the L2 wind product by a gridding tool, nc_L2_to_L3, which interpolates the wind vector cell measurements to a regular lat-lon grid (Figure 2). The measurements from ascending and descending passes are gridded into separate datasets. The gridding procedure is a bilinear interpolation technique within triangles spanned up by observation points and interpolated observations, using Gouraud shading technique. This gridding procedure is applied to all L2 variables, except for the wind vector cell quality flag, which is set when any of the observation points used has a flag set.

Wind speed and wind direction, as well as the zonal and meridional wind vector components are interpolated from L2 to L3. Wind speed calculated from the vector components at L3 may deviate from the wind speed interpolated from L2. Differences are generally negligible (< 0.1 m/s), but can locally be larger where sharp gradients in the wind vector components occur. Users that need consistent values of wind speed and the vector components are recommended to calculate the wind speed from the vector components at L3.

For L3 grid points that are covered by multiple ascending or descending overpasses during a day, the interpolated value from the overpass with the latest timestamp is included in the L3 product.

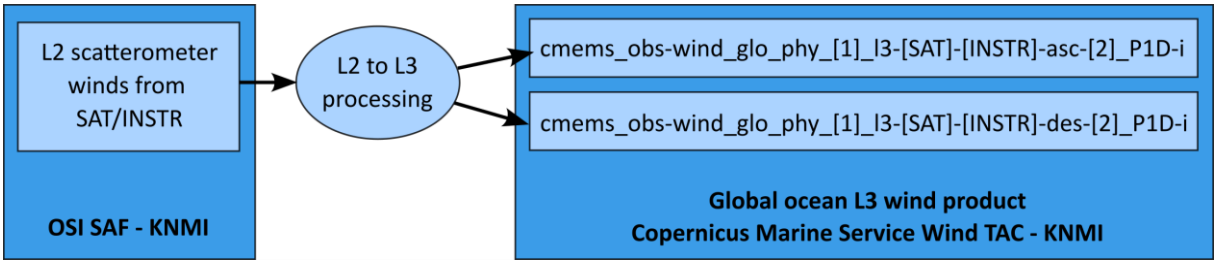

Figure 2: *Schematic view of the L3 processing procedure (nc_L2_to_L3) for scatterometer instrument "INSTR" on satellite "SAT". The ascending (asc) and descending (des) parts of the orbits are separated into different datasets, where [1] is either 'nrt' or 'my' and [2] indicates the horizontal grid.*

II.2 Interpolation to the L3 grid

The gridding procedure is using triangles which are constructed from the centre of each measured Wind Vector Cell (WVC) and two interpolation points (Fig. 3). Any information from WVC's from previous passes is discarded in these triangles.

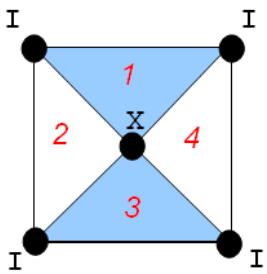

Figure 3*: Schematic view of the triangles used in the L2 to L3 regridding procedure, spanned by the centre of the wind vector cell (X) and the four surrounding interpolation points (I).*

The interpolation points are constructed in the following way to ensure as much coverage as possible:

1. If 4 surrounding WVC are available an interpolation point is constructed in the middle of the WVC's by means of interpolation:

```
6   X   X   X   -   X   X   X          X: measured WVC
5     I   I   -   -   I   I            I: interpolated values
4   X   X   X   X   X   X   X
3     I   I   I   I   I   I
2   X   X   X   X   X   X   X
```

2. Extend swath to the left and fill gaps by extrapolation of two interpolation points to the right:

```
6   X   X   X   -   X   X   X      I: interpolated values
5 I   I   I   I   I   I   I   I          in this step
4   X   X   X   X   X   X   X
3 I   I   I   I   I   I   I   I
2   X   X   X   X   X   X   X
```

3. Extend swath to the right and fill gaps by extrapolation of two interpolation points to the left:

```
6   X   X   X   -   X   X   X      I: interpolated values
5 I   I   I   I   I   I   I   I          in this step
4   X   X   X   X   X   X   X
3 I   I   I   I   I   I   I   I
2   X   X   X   X   X   X   X
```

4. Extend swath before the first row of WVC's and after the last row of WVC's and fill lacking interpolation points within the swath by extrapolation of two interpolation points after, respectively before:

```
7 I   I   I   I   I   I   I   I
6   X   X   X   -   X   X   X      I: interpolated values
5 I   I   I   I   I   I   I   I          in these steps
4   X   X   X   X   X   X   X
3 I   I   I   I   I   I   I   I
2   X   X   X   X   X   X   X
1 I   I   I   I   I   I   I   I
```

5. Fill all interpolation points with wind component u and v values by interpolation from the available surrounding WVC measurements. Only the WVC's that passed KNMI Quality Control are used.

For each constructed triangle, all grid points in that triangle get a *u* and *v* value based on the observation value and the interpolated values at the triangle corners (Figure). For this bilinear interpolation the Gouraud shading technique[1] is used (also known as linear interpolation on barycentric coordinates in a triangle). Gouraud shading can deal with the irregular grids provided by L2 scatterometer products.

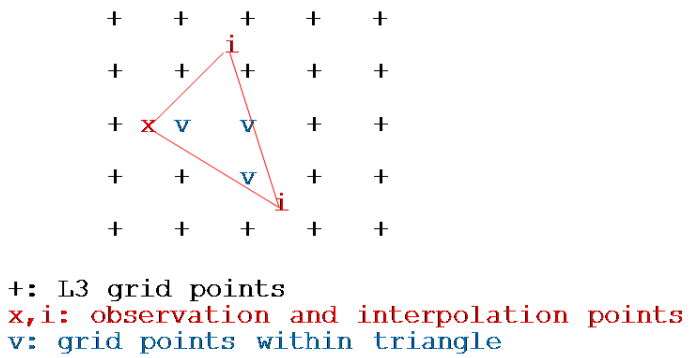

***Figure 4**: Illustration of grid points that are filled with bilinear interpolated values of u and v in the triangle defined by an observation point (WVC centre) and two interpolated observation points.*

[1] See for example: https://codeplea.com/triangular-interpolation.

II.3 L3 wind stress calculation

The backscatter recorded by scatterometers is directly related to ocean wind stress. However, the empirical geophysical model function used in the data processing by definition relates the scatterometer backscatter to the 10 m wind vector and not to the ocean wind stress vector. This choice has been made to allow for accurate verification with buoy observations and model winds.

The surface wind stresses (τ) provided in the products are calculated from the stress-equivalent winds (U_{10s}) according to the standard formula:

$$\tau = C_{D10}\langle\rho\rangle|U_{10s}|U_{10s} \tag{1}$$

In which we use a standard average air density value for $\langle\rho\rangle$ of 1.225 kg/m^3 and a drag coefficient C_{D10} which is determined from the ECMWF ERA Interim model by fitting a line to a full year of data (see Figure 5 and De Kloe et al (2017)). The resulting line is characterised by the equation:

$$C_{D10}(U_{10s}) = aU_{10s} + b \tag{2}$$

In which $a = 7.94 \times 10^{-5}$ and $b = 6.12 \times 10^{-4}$.
This is very close to the relation used by the Coare3.5 (Edson *et al*, 2013) parameterisation.

The function defined above is used to convert both the scatterometer and the model 10m stress-equivalent winds to stress. We use constant values instead of model values for the air density and drag coefficient parameters to avoid any dependence of the scatterometer stress on a particular model. The model air density is given in the product, allowing the user to calculate the stress with his/her preferred formula.

No global reference data are currently available that could be used to derive true Estimated Accuracy Numbers for wind stress. However, as a first estimate we can report the difference between the simple linear approach used in the current L3 wind product and the more complex model used in the ECMWF ERA Interim model. Taking one standard deviation in drag coefficient as error estimate, this can be translated to the error estimates in wind stress as shown in Table 4. The difference and the standard deviation between the L3 stress values and the model values increases with wind speed. The relative error is smallest for wind speeds in the range 5-10 m/s. It should be noted that this estimate is only the result of comparing two models, and may be an underestimate of the actual errors.

***Table 4**: Difference between wind stress (τ) in the current L3 wind product and wind stress in the ERA Interim model for one year of data (2012). An average standard deviation (std) is reported for 5 m/s wind speed bins, and also the relative std. compared to the wind stress itself is given.*

U10s	**0-5 m/s**	**5-10 m/s**	**10-15 m/s**	**15-20 m/s**	**20-25 m/s**	**25-30 m/s**
Avg. τ	0.0062	0.083	0.30	0.75	1.50	2.4
Avg. std τ	0.0006	0.005	0.02	0.07	0.18	0.4
%	9.7	5.5	6.8	9.3	12.0	16.9

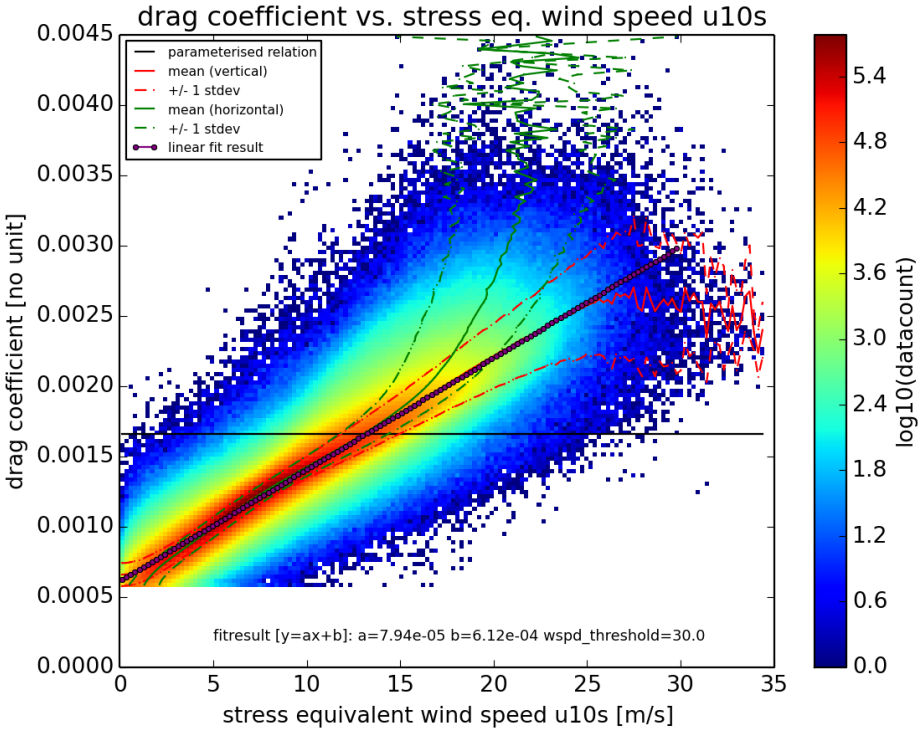

***Figure 5**: Relation between U10s and drag as calculated by the ECMWF ERA Interim wave model for the full year 2012. The colours represent the log10 of the number of datapoints used to calculate each value. Lines show the linear fit (thick black line), the parameterised relation (thin black line), the mean value for every vertical (U10s, green line) and horizontal (drag coefficient, red line) bin and the associated standard deviation (dash-dotted green and red lines respectively).*

II.4 Stress-equivalent wind and equivalent neutral wind

The empirical geophysical model function used to derive the scatterometer wind has no atmospheric information and therefore provides a stress-equivalent wind by definition. To allow for direct comparison, the reference ECMWF model wind provided in the L3 wind products is also a stress-equivalent wind, calculated from the ECMWF 10 meter wind (De Kloe et al, 2017).

The ECMWF 10 meter wind is first converted into an equivalent neutral wind using a stand-alone version of the ECMWF surface layer model. The model air density (ρ) is subsequently used to convert the equivalent neutral wind into a stress-equivalent wind:

$$U_{10s} = \sqrt{\frac{\rho}{\langle\rho\rangle}}\, U_{10,\mathrm{en}} \quad . \tag{3}$$

Users that require equivalent neutral winds instead of stress-equivalent winds can use this formula together with the model air density provided in the product to derive the model and scatterometer equivalent neutral winds.

II.5 Divergence and rotation of the wind and wind stress

Divergence and vorticity (also rotation or curl) of the surface wind and stress field are closely related to vertical motion in the atmosphere and the ocean, such as atmospheric convection and Ekman transport in the upper ocean layers. Horizontal divergence (δ) and rotation (ζ) of the wind field (U) can be calculated from the wind vector components (u and v):

$$\delta = \nabla \cdot \vec{U} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \tag{4}$$

$$\zeta = \nabla \times \vec{U} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \tag{5}$$

Positive divergence values correspond to wind divergence, negative values convergence. Positive vorticity indicates counterclockwise motion, while negative rotation corresponds to clockwise motion (Figure 6).

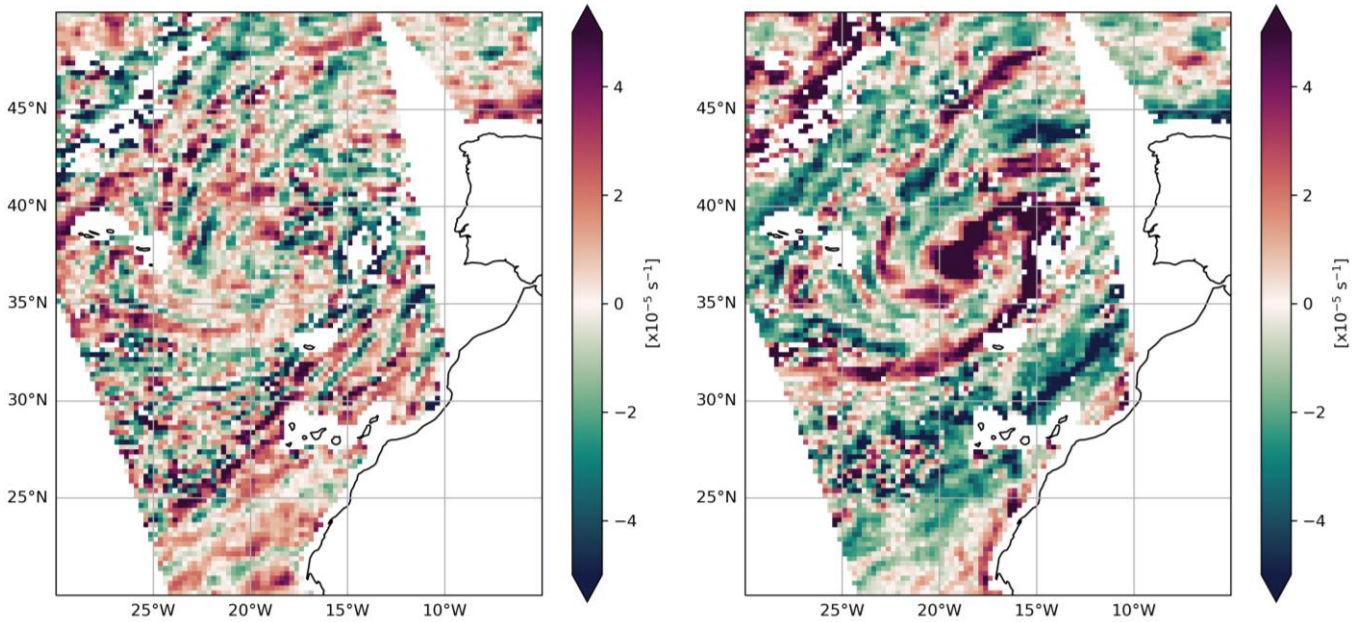

***Figure 6**: Example L3 wind divergence and wind vorticity fields from HY-2B HSCAT on 24 October 2022 (ascending). Spatial derivatives are not calculated for grid cells close to land and ice and with rain present.*

The L3 wind products contain the wind and wind stress divergence and rotation, calculated as spatial derivatives of the L2 wind and wind stress fields. The derivatives are computed on a staggered grid from the average gradients between the four neighbouring wind vector cells (WVC) of the L2 swath grid (Figure 7*). The values on the staggered grid are interpolated to the L3 grid in the same way as the values on the original grid (Section* II.2*). Since scalar quantities are invariant under rotation of the coordinate system, the computed values are also representative for the orientation of the L3 grid.*

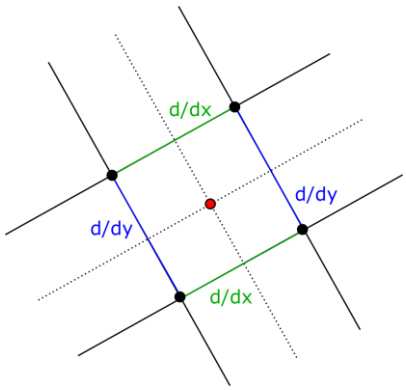

***Figure 7**: Calculation of the spatial derivatives (d/dx and d/dy) for a point (red dot) on a staggered grid (dotted lines) using the four surrounding values (black dots) on the L2 swath grid (solid lines).*

An analysis of wind divergence for different scatterometer instrument types (ASCAT and HSCAT) has shown that the ASCAT wind divergence distribution includes more extreme divergence and convergence values (longer tails) than the HSCAT distribution (Figure 8 (a); Miguel Fernandez, 2023). The wind divergence distributions for the same instrument type have the same characteristics. This also holds for different horizontal grid sizes after normalisation by the standard deviation (Figure 8 (b) and (c)). Wind divergence from the collocated model winds in the L3 datasets (ECMWF operational model) has considerably shorter tails, showing that extreme wind divergence and convergence are not captured in the model. King et al. (2022) show that both extreme convergence and divergence are associated with fast moist convective processes. Such processes are also known to generate extreme shear (vorticity).

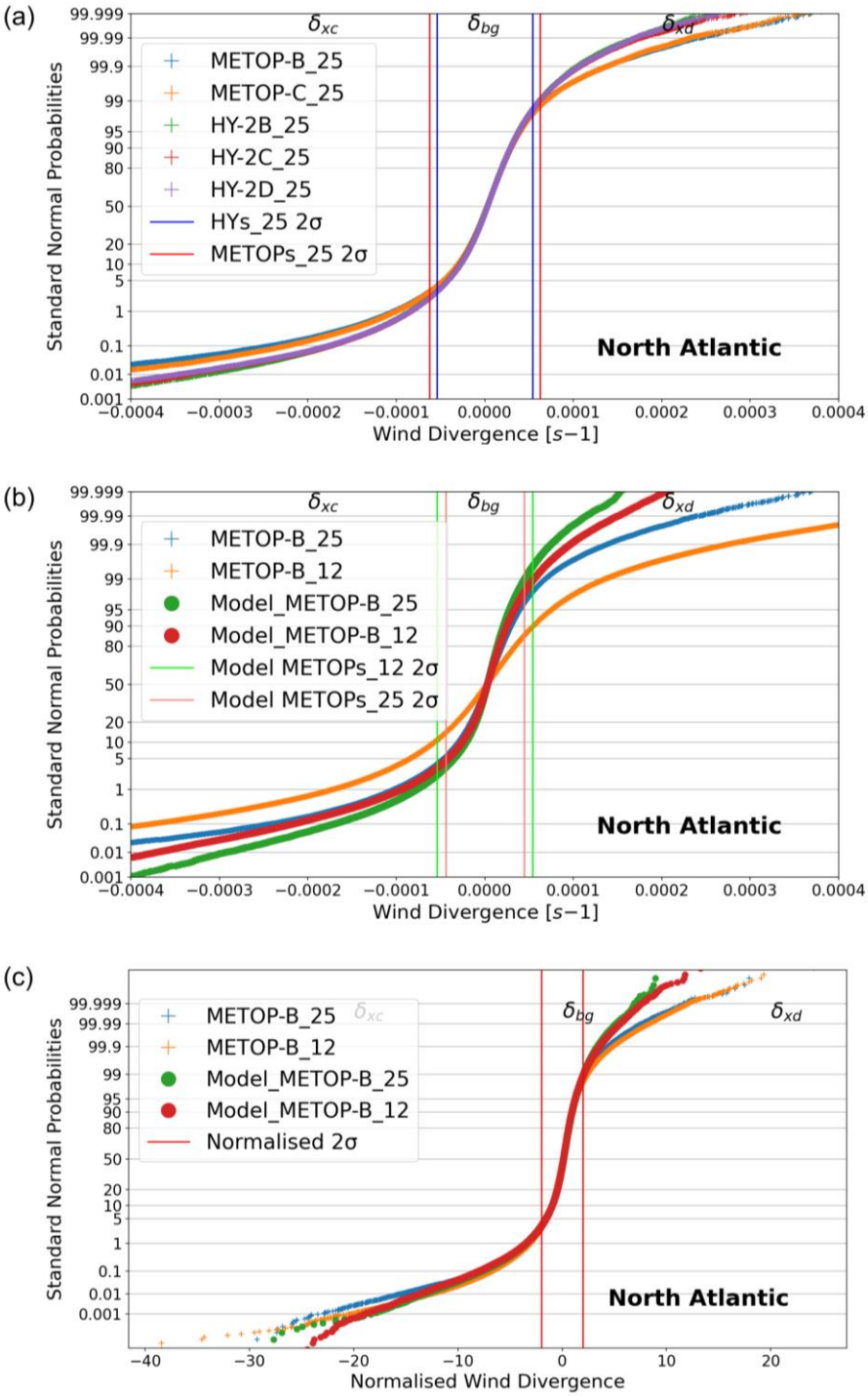

***Figure 8**: Probability distribution of the wind divergence over the North Atlantic (15-70°N) for the period January-March 2022. (a) Comparison of Metop-B and Metop-C ASCAT, HY-2B, HY-2C and HY-2D HSCAT at 0.25° (25) horizontal grid. (b) Comparison of Metop-B ASCAT at two horizontal grid sizes and the collocated model wind divergence. (c) Normalised version of (b). The vertical lines mark the 2 sigma values for the different instruments, dividing the wind divergence range into extreme convergence (δ_{xc}), extreme divergence (δ_{xd}) and background (δ_{bg}).*

II.6 Scatterometer wind validity and quality monitoring

In some cases, scatterometer winds have reduced quality and should be used with care. Examples are extremely high speeds and rainy conditions.

Scatterometer winds are useful (no better alternative) up to 40 m/s. While the in-situ reference scale for calibration of high and extreme winds is controversial, the scales used by KNMI and ECMWF are consistent (EUMETSAT CHEFS project, 2020). To indicate the uncertainty bandwidth, scatterometer winds of 40 m/s produced by KNMI would become 65 m/s winds on a scale used by other producers. However, winds above 40 m/s are very rare at the horizontal resolution of the L3 wind products. The validity range therefore only becomes relevant for applications focused on extreme winds, for example hurricanes.

Rain affects the quality of the wind retrievals, in particular for the higher frequency scatterometers. Rain drops in the radar beam both attenuate and backscatter the microwave signal, reducing the signal contribution returning from the sea surface. For scatterometers operating at Ku-band frequencies (SeaWinds on QuikSCAT, OSCAT on Oceansat-2, Oceansat-3 and ScatSat-1, HSCAT on HY-2B and HY-2C) these effects are strong compared to scatterometers operating at C-band frequencies (SCAT on ERS-1 and -2, ASCAT on Metop-A, -B and -C) (Xu and Stoffelen, 2020). In addition, splashing rain drops stir the water surface and decrease the sea surface roughness, modifying the signal received by scatterometers (Alpers et al., 2016).

In the L2 processing, quality monitoring is performed to flag wind vector cells with possibly degraded wind information, for example due to the local geophysical conditions (rain, sea state) or issues in the wind retrieval procedure. The L2 quality flags are transferred to the L3 regular grid and included in the L3 products. When any of the L2 interpolation points has flags sets, these flags are also set at the interpolated L3 point.

The integer values in the L3 dataset variable 'wvc_quality_flag' represent the sum of the powers of 2 of the flag number (Table 5). For example, when flag numbers 9 (rain detected) and 17 (KNMI quality control fails) are set, the value stored is $2^9 + 2^{17} = 131584$. To determine whether a certain flag is set, the function $\frac{w}{2^f} \bmod 2$ needs to be evaluated, where *w* is the 'wvc_quality_flag' integer value, *f* is the flag number and 'mod' is the modulo operation. The result is 1 when the flag is set and 0 when the flag is not set.

It is recommended not to use grid cells with the variational quality control flag (16), the KNMI quality control flag (17) or the product monitoring flag (18) set. Winds flagged by the rain flag (9) and the ice flag (14) are included in the KNMI quality control flag (17). For the derivative variables (divergence, rotation), the L3 value is set as missing when one or more of the L2 interpolation points has the variational quality control flag or the KNMI quality control flag set. For more information about the flag definitions, see the L2 product manuals for the different instruments.

***Table 5**: Wind vector cell quality flag definitions.*

Flag number	2^(flag number)	Description
6	64	Distance to geophysical model function too large
7	128	Data are redundant
8	256	No meteorological background used
9	512	Rain detected
10	1024	Rain flag not usable
11	2048	Small wind less than or equal to 3 m s^{-1}
12	4096	Large wind greater than 30 m s^{-1}
13	8192	Wind inversion not successful
14	16384	Some portion of wind vector cell is over ice
15	32768	Some portion of wind vector cell is over land
16	65536	Variational quality control fails
17	131072	KNMI quality control fails
18	262144	Product monitoring event flag
19	524288	Product monitoring not used
20	1048576	Any beam noise content above threshold
21	2097152	Poor azimuth diversity
22	4194304	Not enough good σ_0 for wind retrieval

III. VALIDATION FRAMEWORK

Validation is a continuous ongoing activity to characterize accuracy and quality of the delivered wind products. It is mainly based on operational data, but can be supported by campaign data.

Each PU is responsible for validation of their products. The Wind TAC validation activities are mostly based on what is already implemented at the partners' institutes and has been shown to be useful.

Description of validation data and procedures and link to validation results for each product are given in the following sections.

III.1 L3 Global Ocean Wind

The calibration as reported in the Scientific Calibration Report of the Level 3 scatterometer wind products with catalogue product names WIND_GLO_PHY_L3_NRT_012_002 and WIND_GLO_PHY_L3_MY_012_005 consists of two parts:

1. Verification that the products are gridded correctly.
2. Verification that the products are CF-1.6 compliant.

The calibration, validation and quality monitoring of the L2 wind product is done in the framework of the OSI SAF through comprehensive comparisons with hourly winds from moored buoy data and collocated ECMWF winds as obtained from the ECMWF MARS archive (see https://scatterometer.knmi.nl). The data of approximately 140 moored buoys spread over the oceans (most of them in the tropical oceans and near Europe and North America) are used. Table 6 provides the URLs for the near real-time and reprocessed L2 product validation reports. In addition, ASCAT L2 calibration and validation is scientifically described in Vogelzang et al. (2011), and Verspeek et al., Vogelzang et al., Verhoef et al., Portabella et al. (rain), and Belmonte et al. (sea ice), all in the IEEE Transactions on Geoscience and Remote Sensing July 2012.

The correct working of the gridding procedure has been tested by applying the same buoy comparison procedure on the gridded measurements in the L3 netCDF files, using a batch of ASCAT-A NRT 25 km data from January to December 2019 (Table 7). Our analysis characterizes the difference between buoy winds and gridded scatterometer winds (speed, direction, *u*- and *v*-components) in order to compare the bias and standard deviation of the L3 winds relative to the original L2 buoy comparison results. Also the comparison results from a direct collocation between L2 and L3 scatterometer winds are shown (see Figure 10). The effect of the gridding procedure is investigated qualitatively for a typical case with overlapping passes and partly missing data. In addition, the gridding method was tested by inserting artificial inputs to the gridding procedure and comparing input and output fields to an exactly known input function.

Unfortunately, no measurement data is available that can be used to validate the curl and divergence fields. Therefore, the above mentioned interpolation test based on artificial inputs was used to derive estimates of these derivative fields.

Table 6: Scatterometer L2 validation report URLs.

	Instrument	Reference
NRT	Metop-A ASCAT Metop-B ASCAT	https://scatterometer.knmi.nl/publications/pdf/ascat_validation.pdf
NRT	Metop-C ASCAT	https://scatterometer.knmi.nl/publications/pdf/osisaf_cdop3_ss3_valrep_ascat_c_winds_1.3.pdf
NRT	ScatSat-1 OSCAT	https://scatterometer.knmi.nl/publications/pdf/osisaf_cdop3_ss3_valrep_scatsat1_winds.pdf
NRT	Oceansat-3 OSCAT	Not yet available
NRT	HY-2B HSCAT, HY-2C HSCAT, HY-2D HSCAT	https://scatterometer.knmi.nl/publications/pdf/osisaf_cdop3_ss3_svr_hy-2_winds.pdf
MY	ERS-1 SCAT ERS-2 SCAT	https://scatterometer.knmi.nl/publications/pdf/osisaf_cdop2_ss3_svr_ERS_L2_winds_datarecord_1.1.pdf
MY	QuikSCAT SeaWinds	https://scatterometer.knmi.nl/publications/pdf/osisaf_cdop3_ss3_svr_ku-band_wind_datarecords_1.0.pdf
MY	Oceansat-2 OSCAT	https://scatterometer.knmi.nl/publications/pdf/osisaf_cdop3_ss3_svr_ku-band_wind_datarecords_1.0.pdf
MY	Metop-A ASCAT	https://scatterometer.knmi.nl/publications/pdf/ascat_a_cdr_validation.pdf

Table 7: Summary of metrics used in the assessment.

Name of Metric	Description	Supporting metric from OSI-SAF verification	Status
Number of collocations	Number of collocated buoy measurements with L2 and L3 scatterometer measurements		
Bias of wind speed difference	Bias of buoy-L3 wind speed difference	Bias of buoy-L2 wind speed difference	Done
Std.Dev. of wind speed difference	Std.Dev. of buoy-L3 wind speed difference	Std.Dev. of buoy-L2 wind speed difference	Done
Bias of E-W wind component difference	Bias of buoy-L3 east-west (u) wind component difference	Bias of buoy-L2 east-west(u) wind component difference	Done
Std.Dev. of E-W wind component difference	Std.Dev. of buoy-L3 east-west (u) wind component difference	Std.Dev. of buoy-L2 east-west (u) wind component difference	Done
Bias of N-S wind component difference	Bias of buoy-L3 north-south (v) wind component difference	Bias of buoy-L2 north-south (v) wind component difference	Done
Std.Dev. of N-S wind component difference	Std.Dev. of buoy-L3 north-south (v) wind component difference	Std.Dev. of buoy-L2 north-south (v) wind component difference	Done

IV. VALIDATION RESULTS

IV.1 L3 Global Ocean wind

The validation study shows that the L3 wind product processing works correctly and the level of accuracy is very similar to the original L2 accuracy as shown by the buoy collocations. The accuracy level is better than 1 m/s in wind component standard deviation with a bias of less than 0.2 m/s in wind speed (following OSI SAF L2 results). The L3 ASCAT NRT 25-km product meets these requirements for the period considered (January to December 2019).

An example of how the gridding routing deals with overlapping passes is shown in Figure 9.

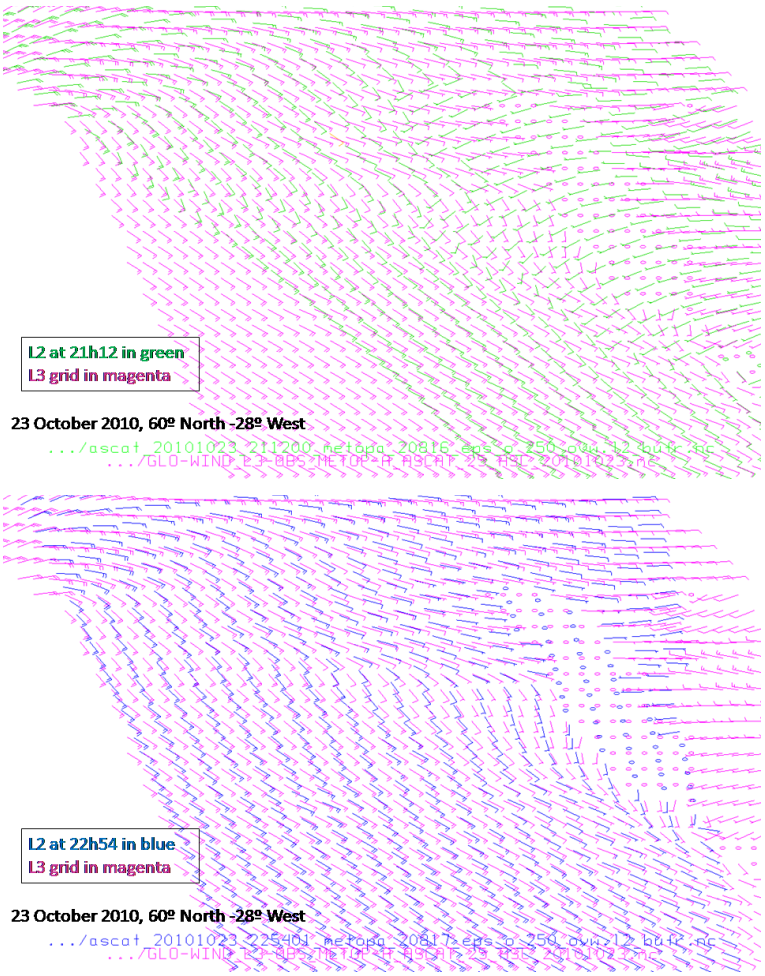

***Figure 9**: The top figure shows the gridding result in magenta and an older pass in green. The bottom figure shows gridding result in magenta and the latest pass data in blue. In the gridding procedure the information from the latest pass is used, and older information is discarded, even if the new pass does not contain valid measurements.*

The comparison of the buoy collocation studies for L2 and L3 observations is summarized in Figure 10 and Figure 11. The L3 wind speed bias follows the monthly fluctuations of the L2 bias. The annual

average of the speed bias is slightly positive for the L3 wind product, while the bias is slightly negative for the L2 product (with a difference of about 0.05 m/s). The difference between the L2 and L3 biases in the *u*- and *v*-components is small as well. The monthly and yearly averages of the standard deviations of speed and *u*- and *v*-components are slightly but systematically larger for the L3 product compared to the L2 product. Note that the largest contribution to the scatterometer-buoy differences is the spatial representativeness error of the buoy for a WVC, where buoy wind vector component errors are typically 1.2 m/s and ASCAT 0.5 m/s (Vogelzang and Stoffelen, 2022). This error will be identical for the L2 and L3 product and hence the additional SD variance in Table 8 may be attributed to the L3 interpolation of scatterometer winds. This implies that the L3 wind component errors are about 50% larger for ASCAT than those of the L2 wind errors. As other scatterometers provide larger errors and smoother winds at L2, these errors are relatively less inflated.

The scatter plots of one year of buoy collocations for L2 and L3 are shown in Figure 12 and Figure 13, respectively. Besides the small mean difference between scatterometer and buoy observations, there is also very good correspondence for individual collocated scatterometer-buoy observations over the entire wind speed and direction range. The resulting average bias and standard deviation of speed, *u* and *v*-components are summarized in Table 8.

Table 8: *Buoy collocation results for L2 and L3 ASCAT-A NRT 25 km wind product over 2019: average bias and standard deviation of speed, u- and v-component.*

2019	bias speed (m/s)	bias *u*-component (m/s)	bias *v*-component (m/s)	std.dev. speed (m/s)	std.dev. *u*-component (m/s)	std.dev. *v*-component (m/s)
L2	-0.02	0.03	0.09	0.98	1.42	1.62
L3	0.03	0.07	0.06	1.11	1.60	1.71

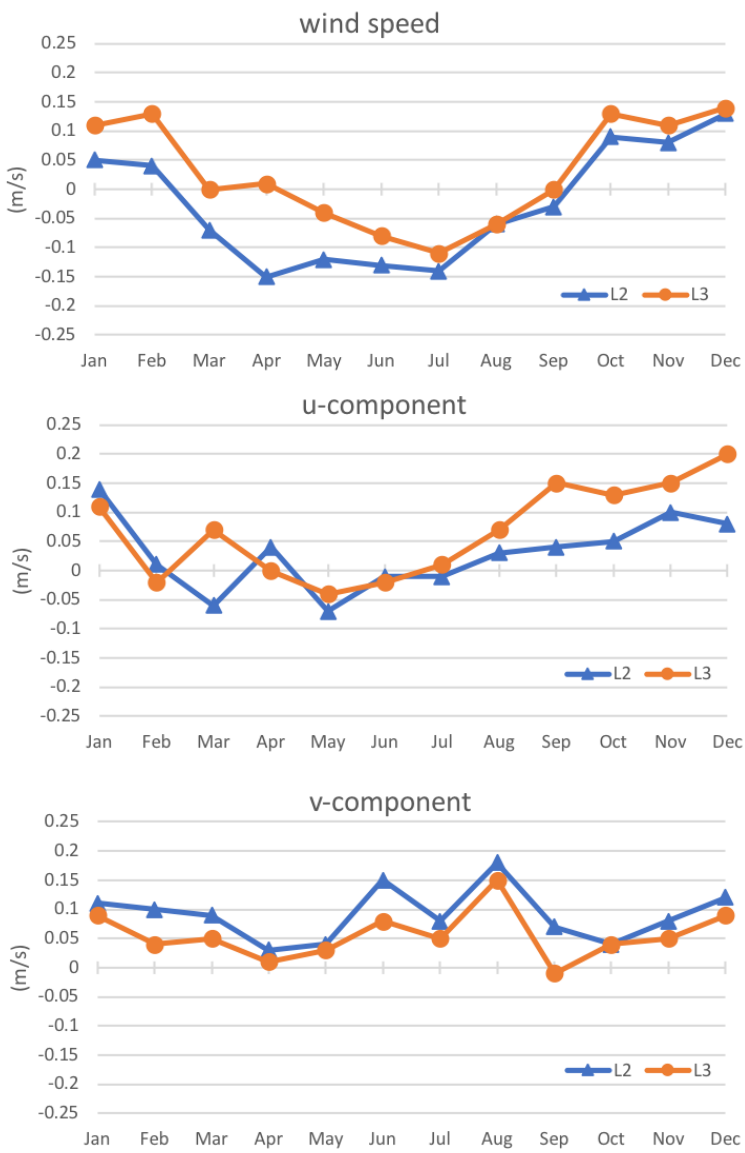

***Figure 10**: BIAS: Buoy collocation results for L2 and L3 ASCAT-A NRT 25 km wind product over 2019 showing the average bias in wind speed (top) and in the u- (middle) and v-components (bottom).*

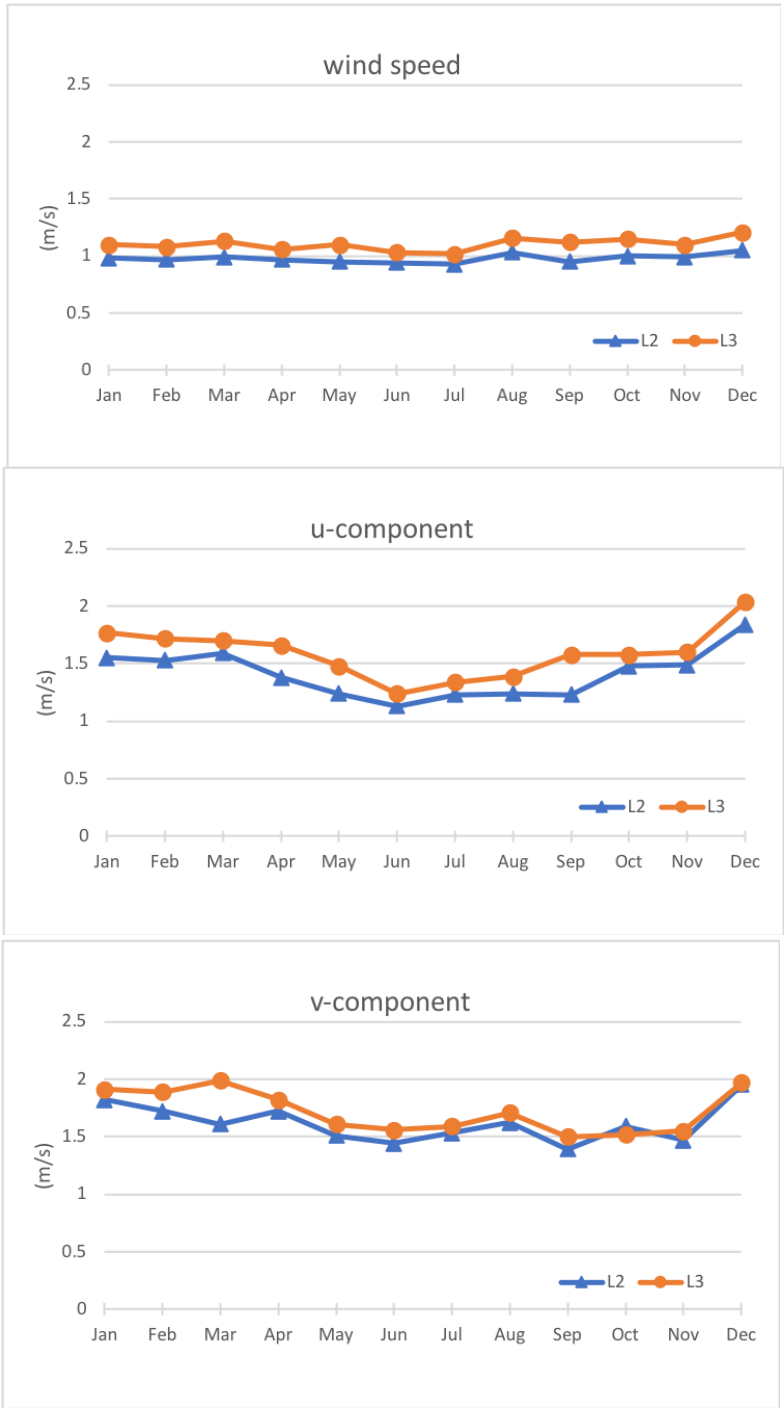

***Figure 11**: STANDARD DEVIATION: Buoy collocation results for L2 and L3 ASCAT-A NRT 25 km wind product over 2019 showing the average standard deviation of wind speed (top) and of the u- (middle) and v-components (bottom).*

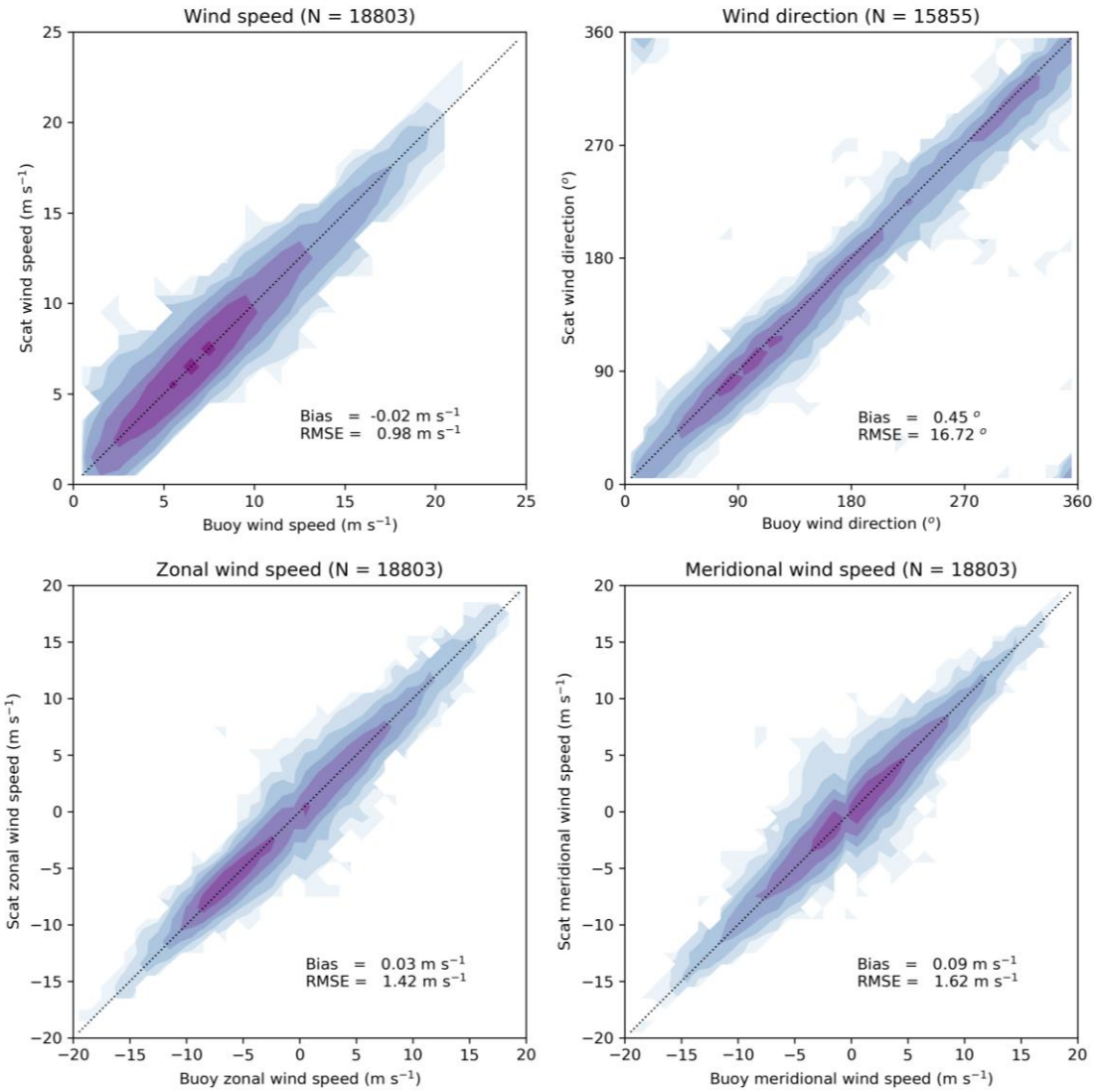

***Figure 12**: Original L2 ASCAT-A NRT 25 km wind product buoy collocation results for wind speed, wind direction and the vector components for the year 2019. N is the number of scatterometer-buoy collocations used and Bias and RMSE the mean bias and root-mean-square difference between scatterometer and buoy observations. The colors illustrate the density of the observations.*

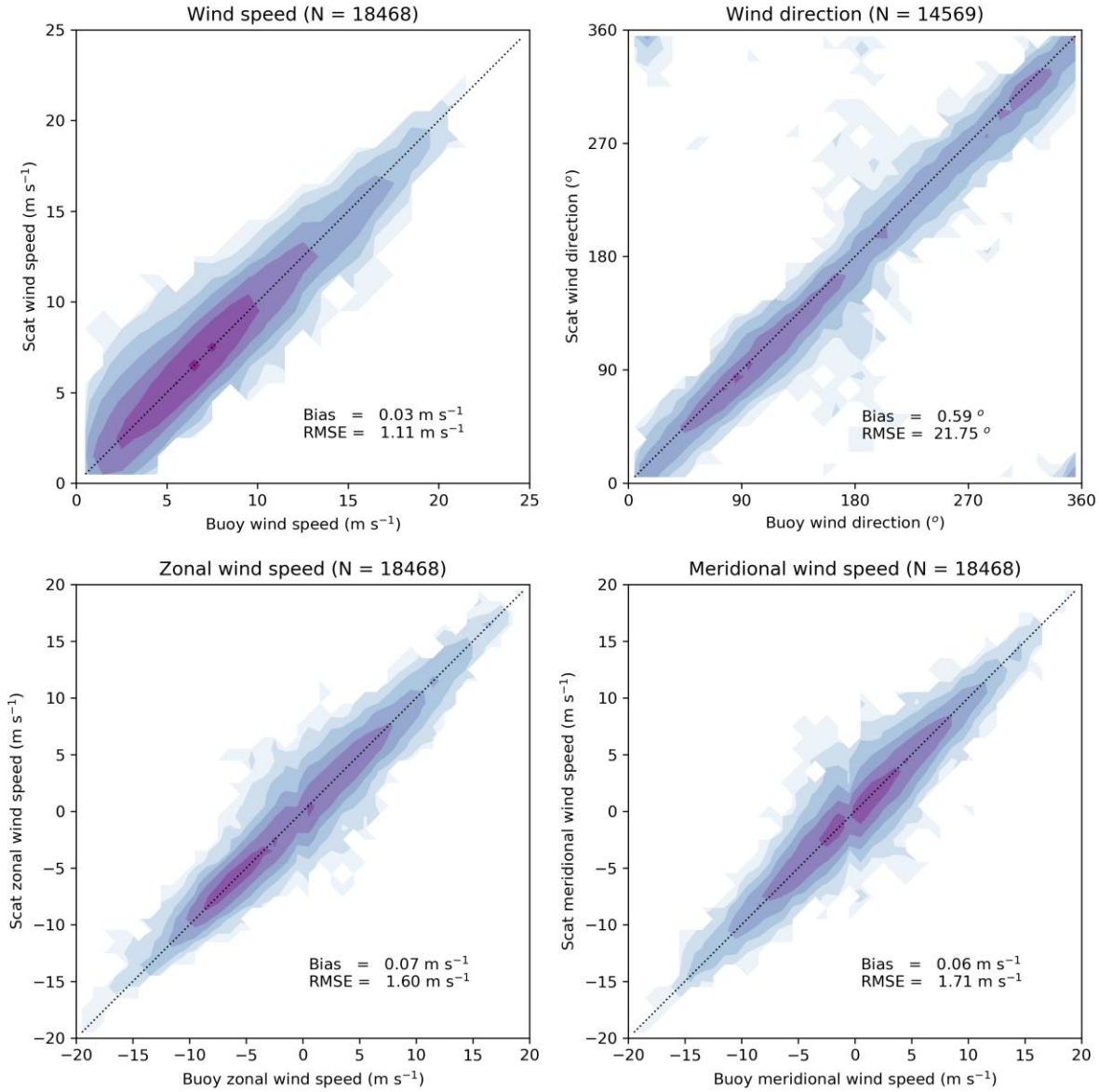

***Figure 13**: Gridded L3 ASCAT-A NRT 25km wind product buoy collocation results for wind speed, wind direction and the vector components for the year 2019. N is the number of scatterometer-buoy collocations used and Bias and RMSE the mean bias and root-mean-square difference between scatterometer and buoy observations. The colors illustrate the density of the observations.*

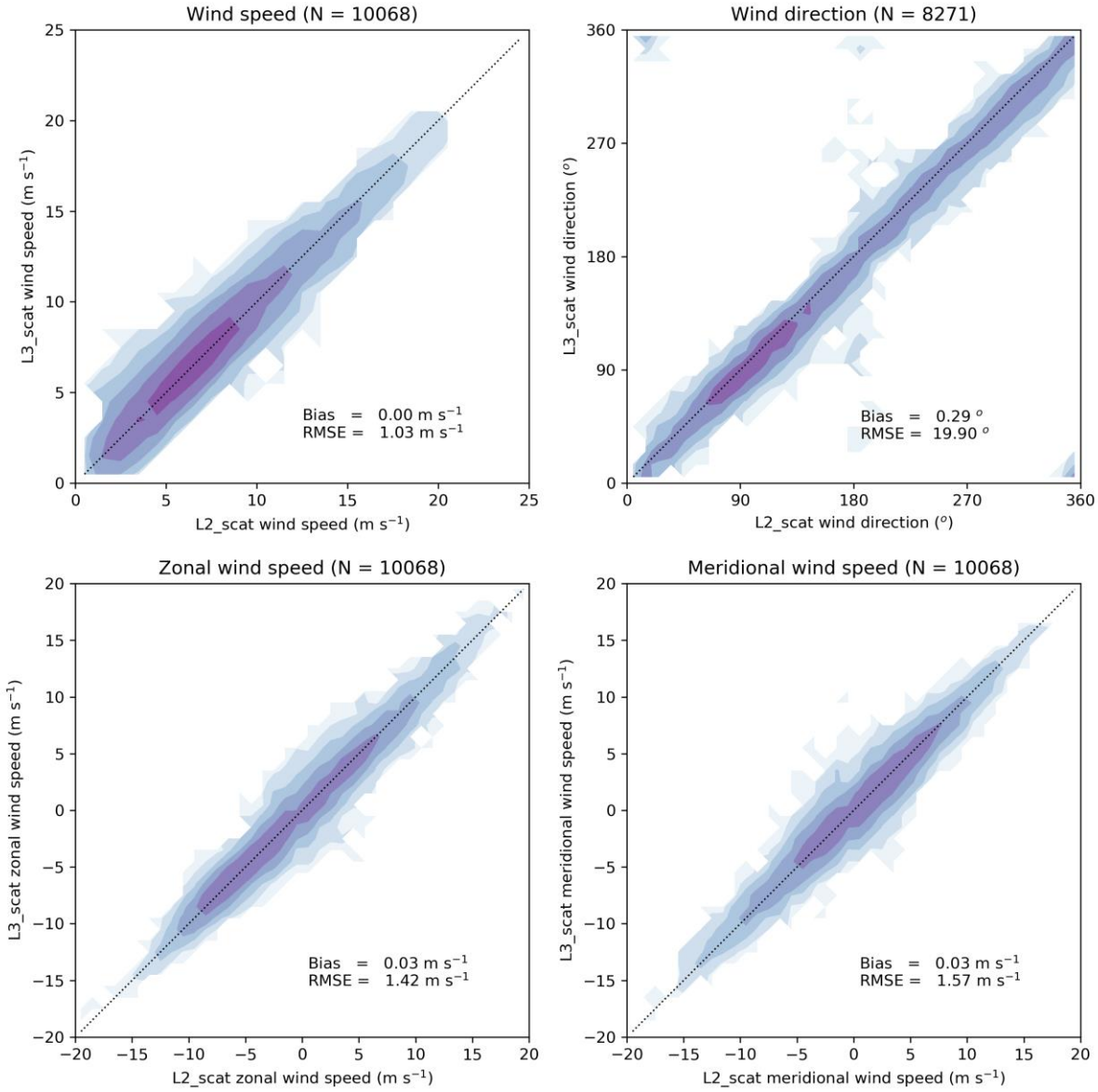

***Figure 14**: Collocation results of L2 and L3 ASCAT-A NRT 25 km wind product for wind speed, wind direction and the vector components for year 2019.N is the number of scatterometer-buoy collocations used and Bias and RMSE the mean bias and root-mean-square difference between scatterometer and buoy observations. The colors illustrate the density of the observations.*

In addition a collocation study between the L2 and L3 product has been performed, using pairs of L2 and L3 measurements with the same buoy measurement (in fact a triple collocation). Figure 14 provides the scatterplots for one year of collocations. The plots show high correlations for the L2 and L3 speed, direction, *u*- and *v*-wind components between 0.99 and 1.0.

Finally, the interpolation from L2 to L3 was tested by inserting artificial inputs to the gridding procedure and comparing input and output fields to an exactly known input function. This procedure follows the following sequence of steps:

- Take an original ASCAT-A L2 BUFR file, read it, select a small orbit section close to the equator, redefine the wind vector using a well know artificial function that depends on latitude and longitude only, and finally write the result back to a valid L2 BUFR file. This procedure also adds invalid wind vector cells on purpose to allow testing the response of the gridding procedure to missing cells.
- For the current test, a 2 dimensional sine variation was chosen, with minimum wind of 0 m/s and maximum wind of 14 m/s. The equation used is:

$$F(x,y) = a + b\ \sin\left(\frac{2\pi x}{s}\right) + b\ \sin\left(\frac{2\pi y}{s}\right) \tag{6}$$

  with *a*=7 m/s, *b*=3.5 m/s, *x*=east-west distance [km], *y*=north-south distance [km], *s*=distance scale [km]. Note that real variation of this size on a 50-km scale are rare, while they are normal on a 1000-km scale, due to the turbulence spectrum. We will therefore evaluate relative errors.

  The derivative in one direction (for example *x*) is:

$$\frac{dF(x,y)}{dx} = \frac{2\pi b}{s}\ \cos\left(\frac{2\pi x}{s}\right) \tag{7}$$

  which has a maximum of $\frac{2\pi b}{s}$.
- Convert the L2 BUFR file to a L2 NetCDF product file, using the standard tool (Bufr2Xmlc) that is also used to generate the NRT and REP products.
- Apply the gridding tool (nc_L2_to_L3) to regrid the artificial data to a regular lat-lon grid and write the result to a L3 NetCDF product file.
- Finally, read the L3 NetCDF product file and compare its content to the function that defines the artificial input winds to the system.

This procedure has been applied using sine functions on latitude and longitude with different scales. An example for a sine with length scale of 1000 km is shown in Figure 15. For each length scale between 50 km and 2000 km, the performance was condensed in a single number by calculating the standard deviation of the detected error over the simulated swath section. The errors depend on the scale of the wind structures, which is as expected. For scales of about 300 km and above, the interpolation error is negligible with less than 10% of the amplitude. Relative errors are larger after the interpolation of small-scale structures, where gradients at the smallest 50 km wavelength scales may be modified by up to 30%.

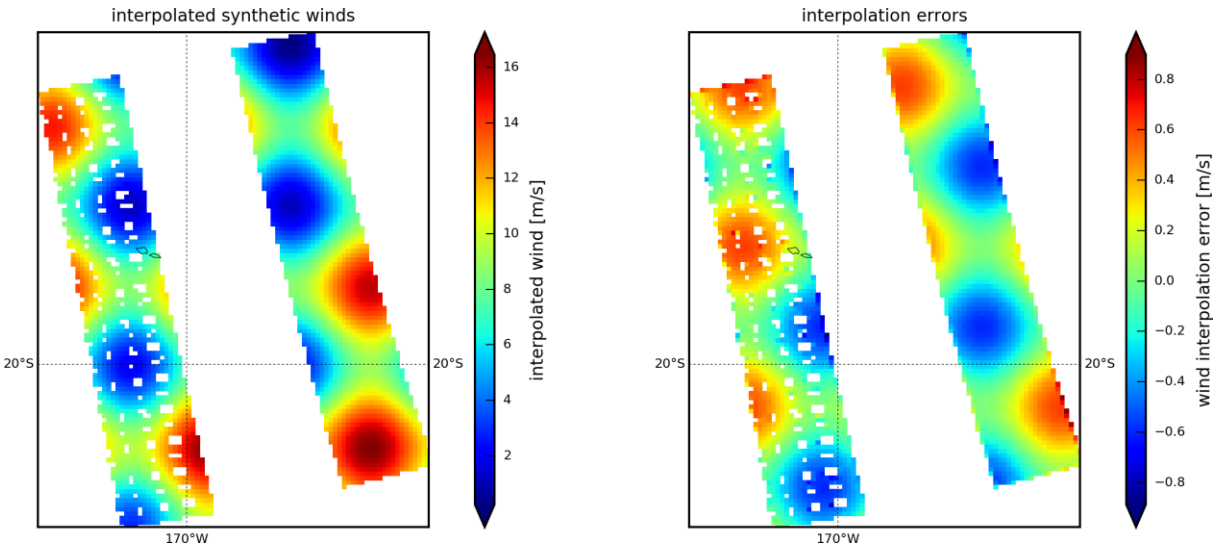

Figure 15: *Example of the results of the interpolation test using artificial input wind speeds (left panel) and the errors in the wind speed field introduced by the interpolation procedure (right panel) for an input sine function with 1000 km length scale along latitude and longitude axes. The white spots in the left swath are cells that have been deliberately flagged invalid to test the response to invalid/missing winds.*

V. SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

V.1 Data gaps resulting from unavailable upstream data

Each L3 datafile contains scatterometer observations over a particular day, aggregated from either all ascending or descending satellite tracks. In cases of satellite anomalies that last longer than a calendar day, no data is available for that day and the datafile is not generated. The files for the day before and after the data gap generally contain a reduced number of observations.

Table 9 lists all missing days in the L3 datasets. As all datasets of a particular instrument are equally affected, they are listed together. For more detailed information about the number of orbits missing, the user is referred to the product pages for the NRT datasets and the L2 product user manuals for the MY datasets, all available via https://scatterometer.knmi.nl/ .

***Table 9**: Data gaps in the L3 NRT and MY datasets (until 1 February 2024)*

Dataset	First day	Last day	Nr of days
cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.125deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-des-0.125deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-des-0.25deg_P1D-i	2017-05-28 2019-07-31	2017-05-28 2019-08-04	1 5 Total: 6
cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-des-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-des-0.5deg_P1D-i	2019-01-04 2019-04-06 2019-04-08 2019-05-05 2019-05-22 2019-08-10 2019-08-31 2019-10-19 2020-03-27 2020-08-08 2020-09-26	2019-01-04 2019-04-06 2019-04-08 2019-05-05 2019-06-24 2019-08-11 2019-09-01 2019-10-19 2020-03-27 2020-08-08 2020-09-26	1 1 1 1 33 2 2 1 1 1 1 Total: 45
cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-des-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat des-0.5deg_P1D-i	2021-10-06 2022-02-22 2022-04-24 2022-05-23 2022-07-09 2023-11-03 2023-11-18 2023-11-25 2023-11-27 2024-01-11 2024-01-20	2021-10-09 2022-02-22 2022-04-24 2022-05-23 2022-07-10 2023-11-03 2023-11-18 2023-11-25 2023-11-27 2024-01-14 2024-01-20	4 1 1 1 2 2 1 1 1 4 1 Total: 19
cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-des-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat des-0.5deg_P1D-i	2023-07-12 2023-11-04 2023-11-18 2023-11-25 2024-01-11 2024-01-20	2023-07-12 2023-11-04 2023-11-18 2023-11-25 2024-01-14 2024-01-20	1 1 1 1 4 1 Total: 9

(This table continues on the next page)

Dataset	First day	Last day	Nr of days
cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-des-0.25deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-asc-0.5deg_P1D-i cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat des-0.5deg_P1D-i	2023-01-20 2023-07-19 2023-10-19 2023-11-03 2023-11-18 2023-11-25 2023-11-27 2024-01-11 2024-01-20 2024-01-27	2023-01-28 2023-07-23 2023-10-19 2023-11-04 2023-11-18 2023-11-25 2023-11-27 2024-01-14 2024-01-20 2024-01-27	9 5 1 2 1 1 1 4 1 1 Total: 26
cmems_obs-wind_glo_phy_my_l3-ers1-scat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-ers1-scat-des-0.25deg_P1D-i	1991-12-11 1992-03-31 1992-06-26 1992-07-20 1993-05-29 1993-06-06 1994-04-01	1991-12-27 1992-04-13 1992-06-28 1992-07-22 1993-05-30 1993-06-06 1994-04-04	17 14 3 3 2 1 4 Total: 44
cmems_obs-wind_glo_phy_my_l3-ers2-scat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-ers2-scat-des-0.25deg_P1D-i	1996-07-16 1998-06-04 2000-01-01 2000-02-08 2000-07-01 2000-10-08	1996-07-25 1998-06-05 2000-01-01 2000-02-09 2000-07-04 2000-10-09	10 2 1 2 4 2 Total: 21
cmems_obs-wind_glo_phy_my_l3-metopa-ascat-asc-0.125deg_P1D-i cmems_obs-wind_glo_phy_my_l3-metopa-ascat-des-0.125deg_P1D-i cmems_obs-wind_glo_phy_my_l3-metopa-ascat-asc-0.25deg_P1D-i cmems_obs-wind_glo_phy_my_l3-metopa-ascat-des-0.25deg_P1D-i	2007-02-28 2007-04-21 2007-09-18 2008-01-17 2008-03-20 2011-05-15 2017-05-28 2019-07-31	2007-02-28 2007-04-24 2007-09-18 2008-01-17 2008-03-20 2011-05-15 2017-05-28 2019-08-04	1 4 1 1 1 1 1 5 Total: 10

(This table continues on the next page)

Dataset	First day	Last day	Nr of days
cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-asc-0.25deg_P1D-i	2010-01-01	2010-01-15	15
cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-des-0.25deg_P1D-i	2010-02-04	2010-02-05	2
cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-asc-0.5deg_P1D-i	2010-02-10	2010-02-11	2
cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-des-0.5deg_P1D-i	2010-03-04	2010-03-05	2
	2010-03-10	2010-03-11	2
	2010-03-15	2010-03-15	1
	2010-04-04	2010-04-05	2
	2010-04-10	2010-04-11	2
	2010-05-04	2010-05-05	2
	2010-05-10	2010-05-11	2
	2010-06-05	2010-06-05	1
	2010-06-10	2010-06-11	2
	2010-07-04	2010-07-05	2
	2010-07-10	2010-07-11	2
	2010-08-03	2010-08-05	3
	2010-08-10	2010-08-11	2
	2010-08-22	2010-08-22	1
	2010-09-04	2010-09-05	2
	2010-09-10	2010-09-11	2
	2010-10-04	2010-10-05	2
	2010-10-10	2010-10-11	2
	2010-11-04	2010-11-05	2
	2010-11-10	2010-11-11	2
	2010-12-04	2010-12-05	2
	2010-12-10	2010-12-11	2
	2011-05-08	2011-05-08	1
	2012-01-01	2012-01-01	1
	2012-02-19	2012-02-19	1
	2012-02-29	2012-02-29	1
	2012-10-02	2012-10-03	2
	2013-03-03	2013-03-04	2
	2013-11-29	2013-12-02	4
	2014-01-01	2014-01-03	3
	2014-02-01	2014-02-03	3
			Total: 79
cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-asc-0.25deg_P1D-i	1999-11-18	1999-11-18	1
cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-des-0.25deg_P1D-i	2000-11-17	2000-11-17	1
cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-asc-0.5deg_P1D-i	2001-05-12	2001-05-13	2
cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-des-0.5deg_P1D-i	2001-07-08	2001-07-08	1
	2001-11-18	2001-11-18	1
	2002-03-20	2002-03-20	1
	2002-11-19	2002-11-19	1
	2006-07-16	2006-07-16	1
	2008-11-26	2008-11-27	2
	2009-09-04	2009-09-04	1
			Total: 12

VI. QUALITY CHANGES SINCE PREVIOUS VERSION

VI.1 Improved coastal coverage for Metop-B and Metop-C ASCAT

As a result of improvements in the upstream processing of the L1 and L2 coastal products for Metop-B and Metop-C ASCAT, scatterometer winds can be resolved closer to the coast (5-10 km). The processing changes also affect winds at a larger distance from the coast. The coastal coverage is improved in the 0.125° NRT datasets for Metop-B and Metop-C ASCAT from 22 November 2022 onwards. Since the NRT Metop-B datasets are used as input for the monthly extensions of the MY product, the 0.125° MY datasets for Metop-B also have improved coastal coverage. Reprocessing of the full Metop-A and Metop-B L2 timeseries with improved coastal coverage is in preparation.

Figure 16 illustrates the increased coastal coverage for the seas around southern Italy. The comparison is based on a test performed with data from Metop-B ASCAT processed for 30 December 2017. The new datasets contain winds for 1-2 grid cells closer to the coast than the original datasets along the entire coastline and islands. The new datasets therefore offer increased possibilities for monitoring and studying processes in coastal regions. The global coverage of the 0.125° datasets increases by 1-1.5%.

Quality changes in the wind fields were assessed by defining all grid cells with quality flags 16, 17 and/or 18 set to fail quality control (following the recommendations in Section II.6). Besides providing increased coverage, the new processing improves the quality of the wind field in coastal regions. Most of the grid cells flags in the original processing are no longer flagged, while many of the newly added grid cells pass quality control. The new processing locally changes zonal wind values by 2 m s^{-1}, including sign changes (Figure 17).

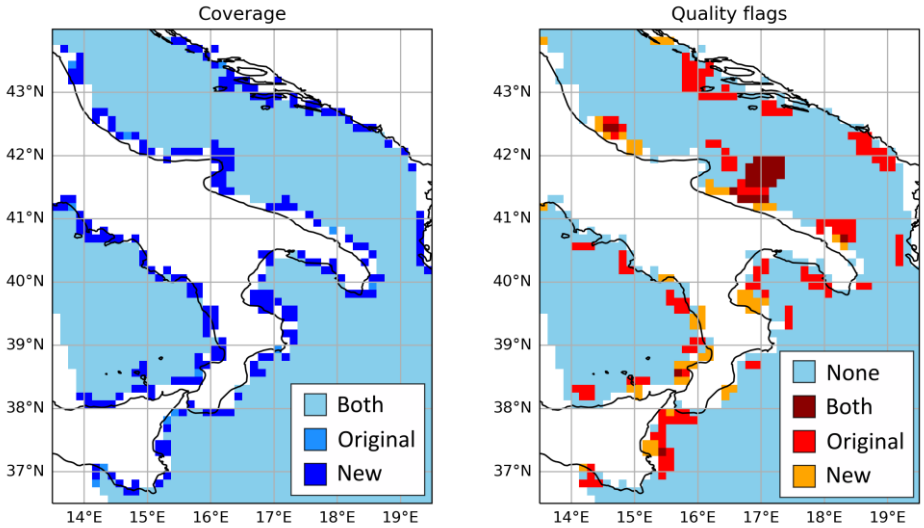

***Figure 16**: Example of changes in coverage and quality control achieved with the new coastal processing, based on a test performed with Metop-B ASCAT datasets for 30 December 2017. The left panel shows grid cells only covered by the new and old processing. The right panel indicates grid cells with quality flags 16, 17 and/or 18 set for only the original or only the new processing and for both.*

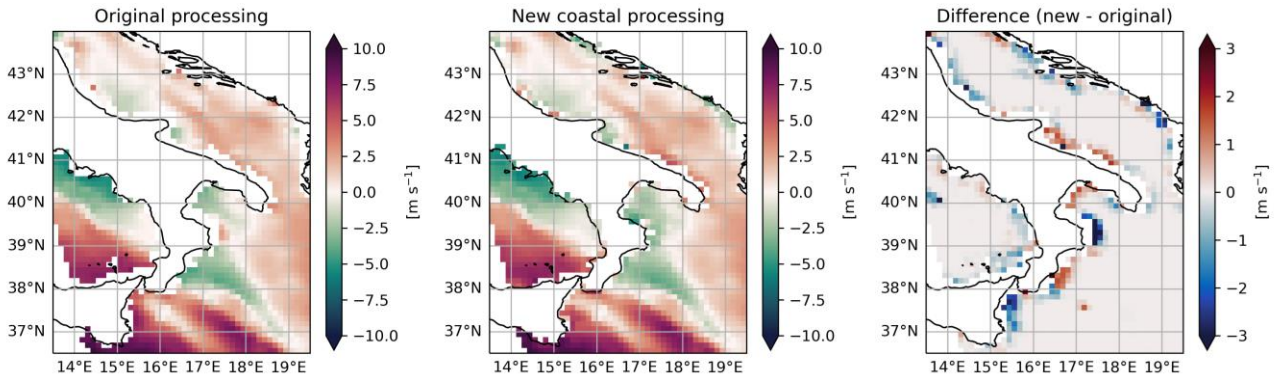

***Figure 17**: Example of changes in coverage and zonal wind values achieved with the new coastal processing, based on a test performed with Metop-B ASCAT datasets for 30 December 2017.*

VI.2 New reprocessed datasets for several scatterometer instruments

Following recent reprocessing of the L2 climate data records for QuikSCAT Seawinds, Oceansat-2 OSCAT, ERS-1 and ERS-2 SCAT in the OSI SAF, the L3 MY datasets for these instruments have been updated. The main changes with respect to the previous datasets are:

- Use of the latest processing software in the OSI SAF L2 processing.
- Use of ECMWF ERA5 background winds instead of ERA-Interim in the L2 processing.
- Collocated ECMWF ERA5 model winds based on hourly instead of 3-hourly forecasts.

The previous and the new reprocessed version for QuikSCAT Seawinds have been compared to a reference dataset (Metop-A ASCAT) over the year 2008 (Figure 18). To reduce the sampling effects, the comparison is based on the annual mean difference between scatterometer observations and collocated ERA5 model winds for all datasets. Differences between meridional wind biases for the two instruments are significantly reduced in the new version, particularly in the tropics and the trade wind region.

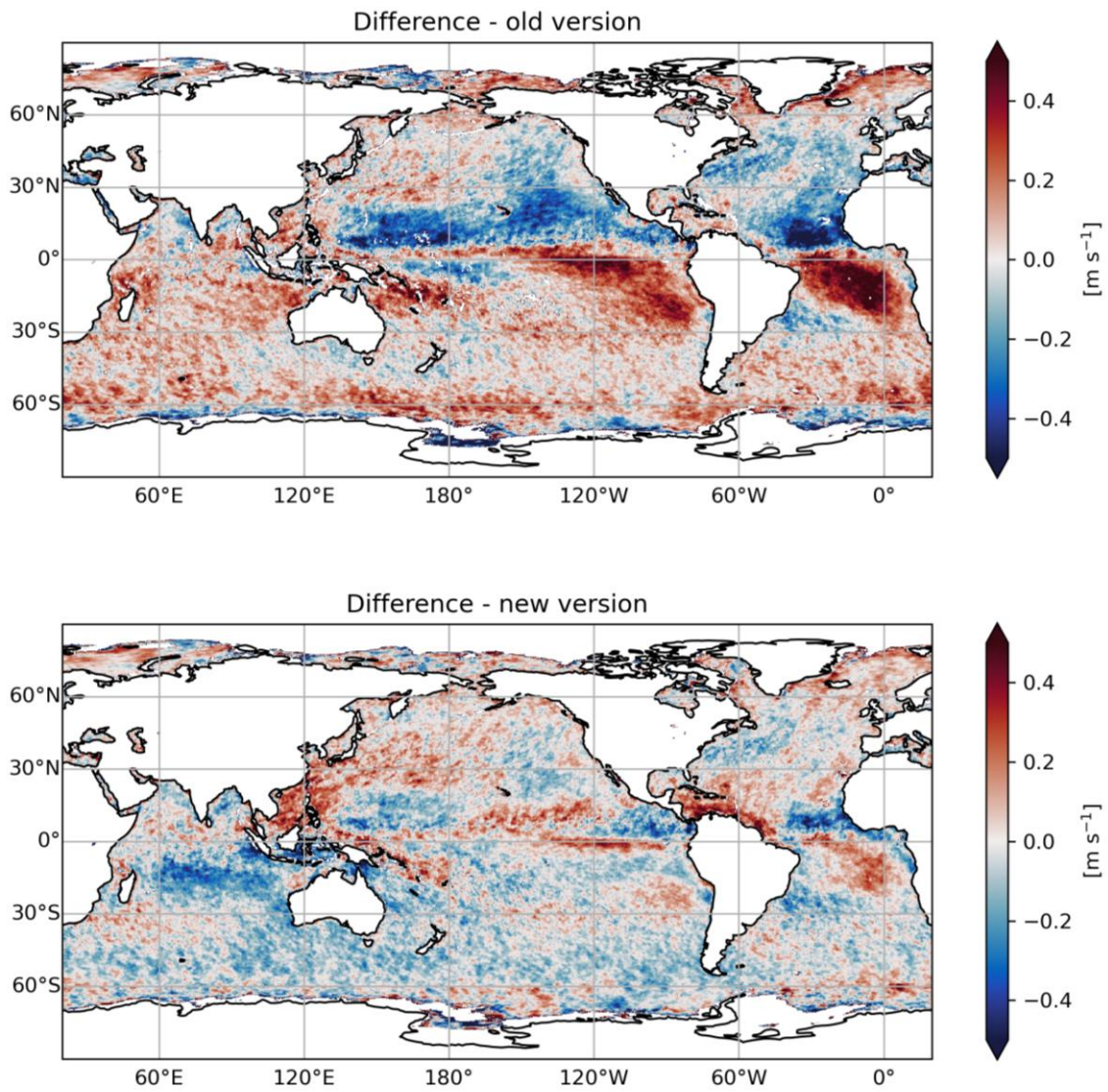

***Figure 18**: Difference between the QuikSCAT Seawinds and Metop-A ASCAT meridional wind component bias (scatterometer-ERA5) averaged over the year 2008 for the old and the newly reprocessed version of the QuikSCAT dataset.*

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